

1. List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

ANS:-

Hardware Issue	Typical Symptom
Defective / Unseated RAM	PC turns on with fans spinning but fails POST, gives continuous beep codes, or shows a black screen.
GPU / Graphics Card Failure	Screen shows visual artifacts, horizontal lines, color distortion, or “No Signal.”
Overheating CPU / Dried Thermal Paste	PC randomly shuts down or freezes during high-load tasks such as gaming or rendering.
Failing Storage Drive (HDD/SSD)	Very slow file transfers, “Boot device not found” errors, or clicking sounds from an HDD.
Degraded Power Supply Unit (PSU)	PC does not power on, or suddenly reboots without displaying an error message.

2. Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

ANS:-

Laptop Not Charging, Troubleshooting Plan:

- Check the Power Outlet: Test another device to make sure the outlet is working.
- Inspect the Charger: Check the adapter and cable for cuts, damage, or loose connections.
- Check the Charging Port: Look for dust, dirt, or a loose connection in the laptop port.
- Test with a Multimeter: Check the charger’s output voltage to confirm it is working properly.
- Check the Battery: If the charger works but the laptop still does not charge, the battery may be faulty.

3. Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

ANS:-

USB Phone Charger Voltage Test

- **Measured Voltage:** 5.12 V
- **Expected Voltage Range:** 4.75 V – 5.25 V (Standard 5V USB tolerance)
- **Result / Conclusion:** The measured output of 5.12 V falls comfortably within the standard operational range. This confirms the USB charger is delivering stable, safe power for mobile devices.

4. A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate? Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires.

ANS:-

Testing the 12V Rail of an ATX Power Supply:

- Set the multimeter to DC voltage (V=) mode.
- Connect the probes: red probe to the yellow wire (+12V) and black probe to the black wire (GND).
- Record the reading: 12.08 V.

Result: The voltage is within the normal range. A reading below 11.5 V may indicate a weak or faulty power supply, which can cause random shutdowns.